

Heat diffusion — modes and decay

A uniform rod, zero-temperature ends and two initial Fourier modes.

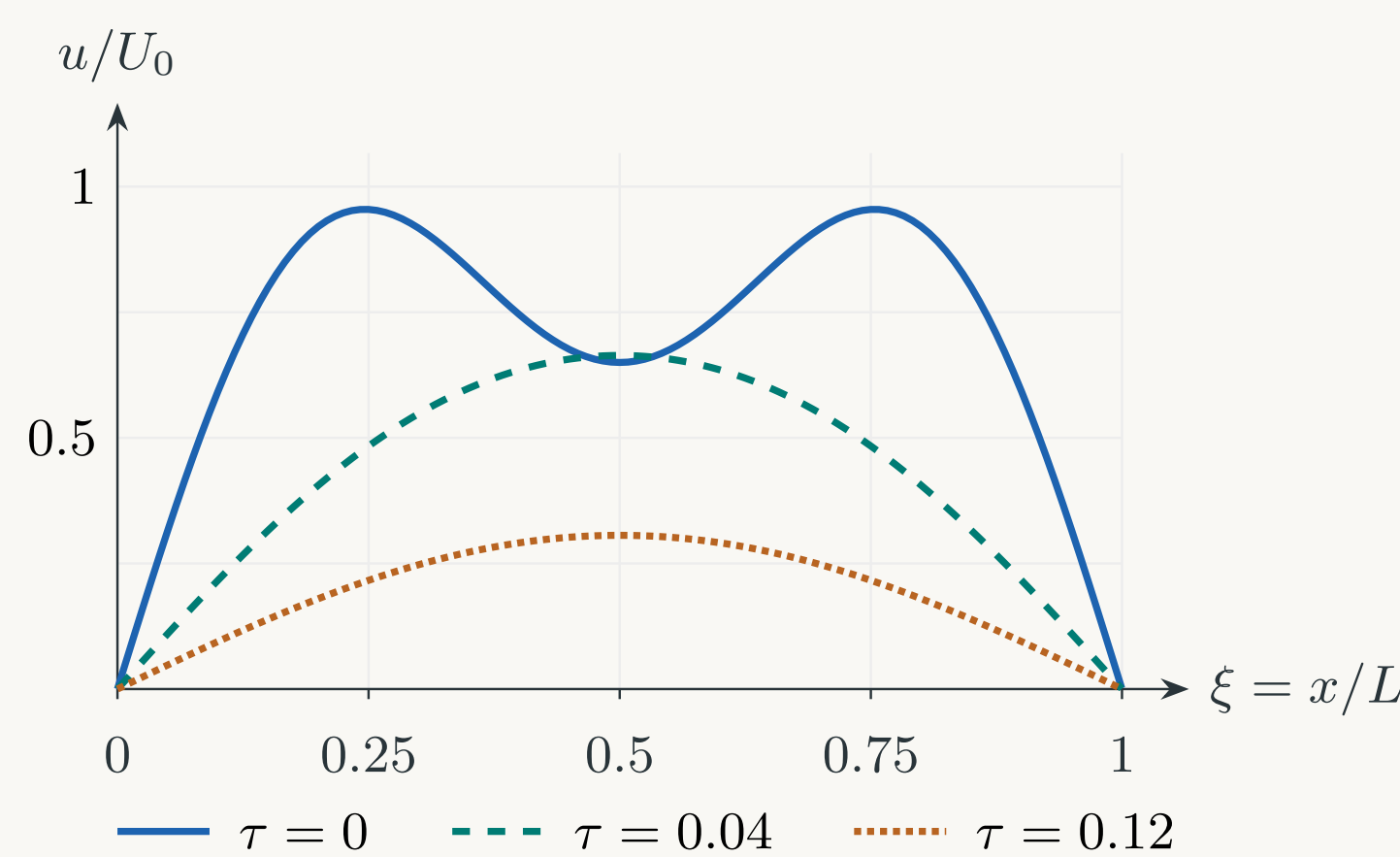
01 / MODEL + CONDITIONS

$$\begin{aligned} \partial_t u &= \kappa \partial_{xx} u, & 0 < x < L, \quad t > 0, \\ u(0, t) &= u(L, t) = 0, & \kappa > 0, \\ u(x, 0) &= U_0 \left[\sin \frac{\pi x}{L} + 0.35 \sin \frac{3\pi x}{L} \right], \\ \xi &= x/L, \quad \tau = \kappa t/L^2, & v &= u/U_0, \\ v_\tau &= v_{\xi\xi}, & 0 < \xi < 1. \end{aligned}$$

02 / EXACT SOLUTION + CHECK

$$\begin{aligned} X_n(x) &= \sin(n\pi x/L), & \lambda_n &= (n\pi/L)^2, \\ T_n(t) &= \exp(-\kappa\lambda_n t), \\ v(\xi, \tau) &= e^{-\pi^2\tau} \sin(\pi\xi) + 0.35e^{-9\pi^2\tau} \sin(3\pi\xi), \\ E(t) &= \tfrac{1}{2} \int_0^L u^2 dx, \\ \frac{dE}{dt} &= -\kappa \int_0^L (u_x)^2 dx \leq 0. \end{aligned}$$

03 / GEOMETRY + BEHAVIOUR



Exact modal solution: the third mode decays nine times faster.

04 / DISCRETISATION

$$\begin{aligned} x_j &= j\Delta x, \quad t_n = n\Delta t, & r &= \frac{\kappa\Delta t}{(\Delta x)^2}, \\ u_j^{n+1} &= u_j^n + r(u_{j-1}^n - 2u_j^n + u_{j+1}^n), \\ 0 \leq r &\leq \tfrac{1}{2} & (\text{explicit stability}), \\ u_0^n &= u_N^n = 0, \quad \Delta x = L/N, \\ \text{error} &= O(\Delta t + \Delta x^2) & \text{for smooth solutions.} \end{aligned}$$

The third mode decays nine times faster. Check the boundary values and energy dissipation.

Select and double-click any equation or diagram to edit its source. Scroll to work below.